

How can I apply transforms.Normalize on DataLoader

DeepLearner17

Jan 2018

Hello,

I have dataset got from numpy. I would like to apply transform.Normalize

```
normalize = transforms.Normalize(mean=[0.485, 0.456, 0.406],
std=[0.229, 0.224, 0.225])
```

which is basically applied on ImageFolder as follow :

```
train_loader = torch.utils.data.DataLoader(
    datasets.ImageFolder(trainindir, transforms.Compose([
        transforms.RandomSizedCrop(224),
        transforms.RandomHorizontalFlip(),
        transforms.ToTensor(),
        normalize,
    ])),
```

However I have numpy dataset so I have access to my dataset as follow :

```
data_train = torch.from_numpy(data_train).expand(-1,3,-1,-1)
labels_train=torch.from_numpy(output_le[:len(labels_train)])
train_data = torch.utils.data.TensorDataset(data_train.float(), labels_train)

train_loader = torch.utils.data.DataLoader(
    train_data, batch_size=args.batch_size, shuffle=(train_sampler is None),
    num_workers=args.workers, pin_memory=True, sampler=train_sampler)
```

How can I apply

```
normalize = transforms.Normalize(mean=[0.485, 0.456, 0.406],
std=[0.229, 0.224, 0.225])
```

In this case ?

Thank you,

simopal6 Simone

Jan 2018

I'm not sure you can apply a transform on `DataLoader` . Maybe you could subclass `TensorDataset` and add a transform argument to the constructor, then override `__getitem__` to call the parent's `__getitem__` and apply the transform to the returned data.

Or, in your case, why don't you just normalize the data before passing it to `TensorDataset` ?

@simopal6 thank you. Yes , I think it is the simplest way to do that.
I'm also wandering why there are 4 values for means and 4 values for std ?

```
mean=[0.485, 0.456, 0.406]
std=[0.229, 0.224, 0.225]
```

Thank you

It's actually 3 and 3 😊 It's because each color channel in the input image is normalized according to its own mean and standard deviation.

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